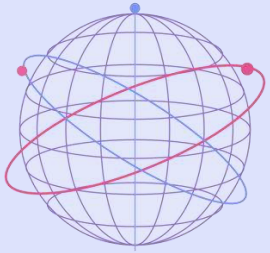
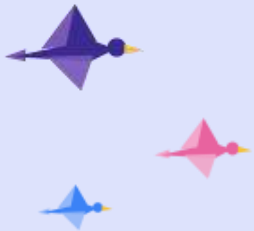
PARTICIPANT PRESENTATION

Qiskit Fall Fest 2026

Hackathon · Team Details





Participant Details

PROJECT TITLE
AAROGYAM: Quantum Assisted Early Health Risk Intelligence

TRACK – 7th
Quantum technology for medical application and sensing

TEAM NAME
HackLords

TEAM LEADER
Pranali Basarikatti

EMAIL / CONTACT
pranalib2026@gmail.com / 9663667098

TEAM MEMBERS
Pratham W · Shreya Bhadrashetti

INSTITUTION / ORGANISATION
Jain College of Engineering and research

COUNTRY
India

Problem Statement

THE PROBLEM

- Risk numbers are hard for people to read
- Clinicians need clear, organised summaries
- Quantum-ML demos rarely show honest tests

THE GAP

Understandable, measurable, responsible

AAROGYAM'S ANSWER

- Plain-language, clearly labelled estimates
- Patient and clinician portals
- Measured classical vs quantum results

Objectives & Novelty

OBJECTIVES

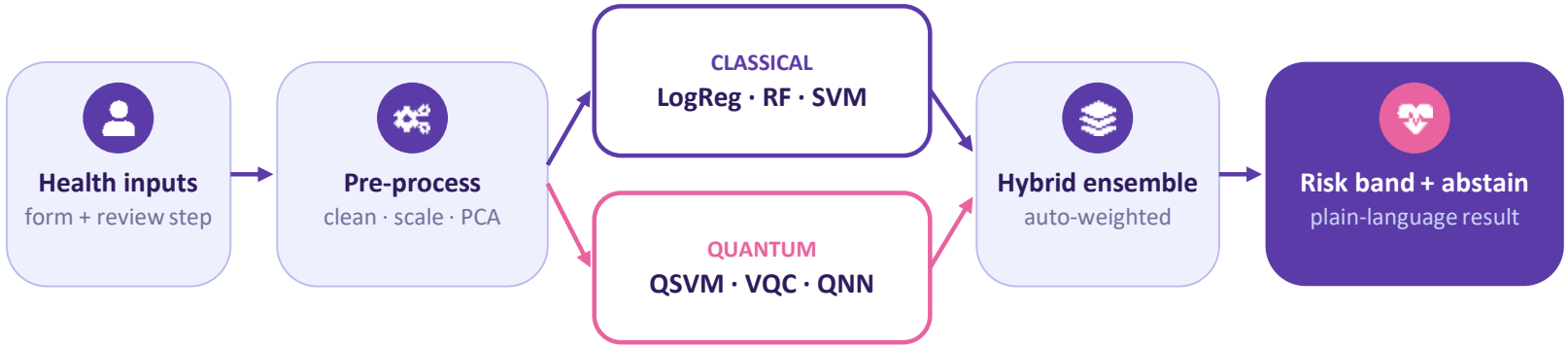
- Two separate tasks: heart and diabetes
- Fair classical vs quantum comparison
- Patient and clinician portals
- Safe by design: QR, dialer, disclaimers

AAROGYAM
+ QGene engine

NOVELTY (QGENE ENGINE)

- Quantum-readiness check before training
- Closed-form kernel: O(n), not $O(n^2)$
- 3 quantum models: QSVM · VQC · QNN
- Conformal abstention + risk bands

Solution & Quantum Advantage



Quantum advantage is tested, not assumed

Quantum is the best single model on 2 of 5 datasets, heart disease included; the ensemble down-weights it elsewhere

Product Experience

PATIENT	**Pick a task** heart or diabetes	**Enter + review** validated form	**Get estimate** risk band, labelled	**Wellness tips** general, no treatment	**Share by QR** opaque token only
CLINICIAN	**Open record** synthetic demo data	**Review inputs** and model result	**See the limits** metrics + caveats	**Scan QR** consent required	**Discuss** aid only, no diagnosis

DEMO DATA

EXPERIMENTAL RESULT

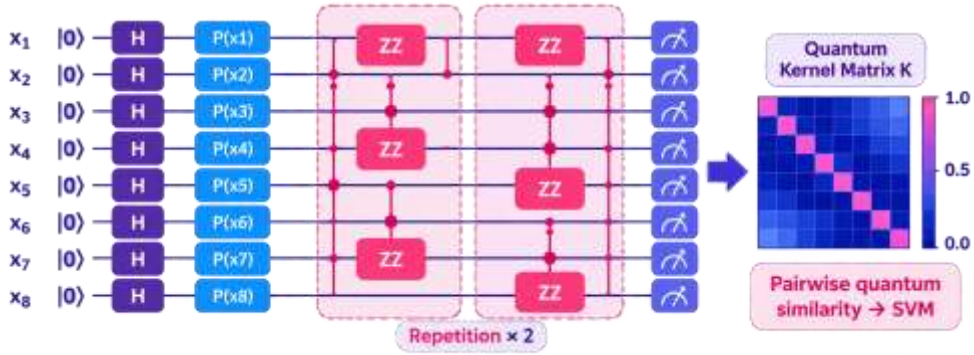
NOT A DIAGNOSIS

Emergency dialer: opens phone, you confirm

Algorithm & Circuit Design

1 Quantum Circuit – ZZFeatureMap (8 qubits)

Feature encoding with single-qubit rotations and ZZ entanglement (reps = 2)



$$K(x, y) = |\langle \varphi(x) \mid \varphi(y) \rangle|^2$$ fidelity between encoded states

Hybrid pipeline

1. Clean · impute · denoise
2. Normalise (train-fit only)
3. PCA → 8 components
4. Rescale to [0, π]
5. ZZFeatureMap → 8 qubits

Statevector kernel: states built once, a 4000×4000 Gram matrix in about 1 s

Implementation & Tech Stack

Experience	React + Vite	React Three Fiber	QR + dialer	Recharts
API	Python · FastAPI	Input validation	/predict · /metrics	/records · /qr
ML engine	scikit-learn	Qiskit 2.x	Closed-form kernel	SPSA · COBYLA
Data	Heart (Cleveland)	Diabetes (Pima)	Synthetic demo records	Upload your CSV
Ship	Docker	Render · Fly.io	Python 3.12	Pinned dependencies

Results & Analysis

Classical vs Quantum — ROC-AUC (%)

Higher is better. Comparison of best classical and best quantum model on each dataset.



Evaluation: held-out test sets | Train : Test = 80 : 20 | **Metric:** ROC-AUC (%)
Models: Best classical (e.g., SVM/XGBoost) vs best quantum (QSVM + ZZFeatureMap)
Note: Research benchmark results on public datasets. Not clinically validated.

Key takeaways

2 / 5
datasets where quantum is the best single model

Heart (CAD)
94.9%
(Quantum)
+2.1 pp AUC

Heart AUC: QNN/QSVM outperforms the best classical (92.8%).

Diabetes
79.4%
(Classical)
+3.2 pp AUC

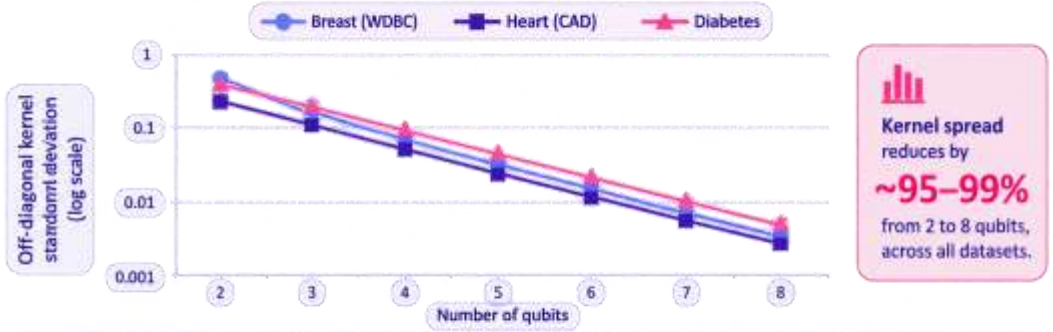
Diabetes: classical model stays ahead of quantum (76.2%).

Overall insight
Quantum models show state-dependent performance. They excel on some datasets (e.g. Heart), but classical models remain stronger on others.

Why Quantum Wins, or Loses

Kernel Concentration vs Qubit Count

Off-diagonal similarity spread decreases as qubits are added



Kernel spread reduces by **~95–99%** from 2 to 8 qubits, across all datasets.

Configuration: ZZFeatureMap | **Qubits:** 2 – 8 | **Repetitions:** 2 | **Metric:** Off-diagonal std | **Simulator:** Statevector (local)

Observation
Off-diagonal kernel spread decreases consistently with more qubits.

Reason
States become more similar (kernel concentration), reducing discrimination.

Implication
Quantum models like QSVM can struggle when the kernel loses useful variation.

CAUSE → EFFECT → IMPACT → RESPONSE

More qubits added
Higher-dimensional feature encoding with ZZ entanglement

Kernel concentration
Off-diagonal similarity spread decreases roughly with each additional qubit

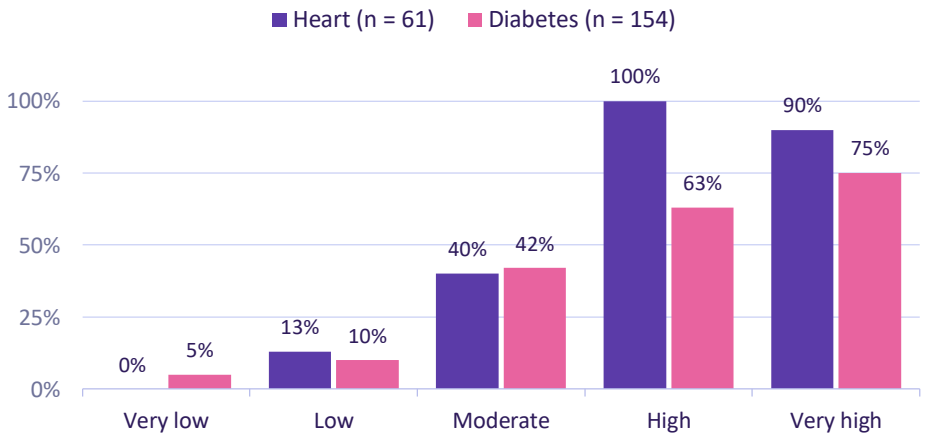
Reduced kernel discrimination
Kernel tends to a near-identity structure, limiting class separability

Observed limitation
Heart QSVM: 7% sensitivity (recall) at the evaluated threshold

AAROGYAM's response
Measure kernel quality before training and adapt ensemble weights to reduce quantum contribution when performance is poor.

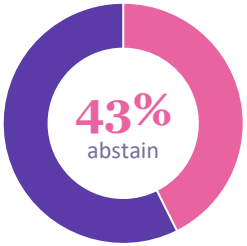
Decision Support

Observed outcome rate in each risk band

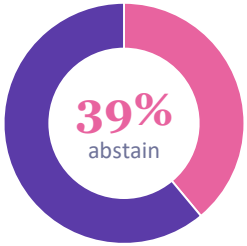


Observed rates mostly rise with band · small held-out test sets

Conformal prediction: commit or abstain



Heart
97% accurate
when it commits



Diabetes
82% accurate
when it commits

Calibration sets are small, so guarantees are loose

Impact & Real-World Applications



Challenges & Future Scope

CHALLENGE	HOW WE HANDLE IT
Kernel concentration	Diagnose it before training
Tiny public datasets	Conformal abstention, honest labels
Public data ≠ real patients	Synthetic demo records, no claims
Slow quantum simulation	Closed-form O(n) kernel

ROADMAP

1. **MVP**
   Dual portals, honest benchmarks, synthetic records
2. **NEXT**
   Authentication, consent, audit trail, clinician review
3. **LATER**
   Real IBM quantum hardware, error mitigation, more diseases

CLOSING

Thank You

Questions?

Inspiring Minds. Shaping Tomorrow.

GITHUB REPOSITORY
Click here

DEMO VIDEO LINK
Click here

